



Analyzing the interaction of critical success factor for TQM implementation- A grey-DEMATEL approach

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Abstract

In an organization, successful implementation of Total Quality Management (TQM) brings lot of benefits in terms of quality improvement, customer satisfaction, profit, and market share. It also assists businesses to improve in terms of performance, reducing rework in the process, and reduction in overall costs associated with poor quality (e.g. replacement, warranty, late deliveries, rework and scrap.). However, in practice many enterprises fail to adopt and implement TQM besides having known its benefits and advantages into the organization. Consequently, there is essential requirement for a further systematic and deeper assessment of the various TQM factors for its successful implementation. By considering this research gap in the context of Indian industry, an exhaustive and planned work was carried out to meet this requirement. Several factors influencing the implementation of TQM were decided by consulting industrial and academic experts. From the available responses, an attempt was made to uncover the causal relationships among the identified critical success factors (CSF's) and categorize them into cause-and-effect groups. In the present work, Grey Set Theory has been integrated with DEMATEL (Decision making Trial and Evaluation Method). Further, the integrated grey based DEMATEL approach was applied to reveal the interrelation among the identified critical success factor. The results indicate that among the identified 15 CSF's for the successful TQM implementation in Indian SMES industries 'Top management commitment (CSF1)', 'Training and education (CSF6)', 'Human resource management (CSF3)' and 'Information system (CSF11)' are topmost influential cause group factor. Whereas 'Process management (CSF4)', 'Customer focus (CSF5)', 'Supply development (CSF7)' and 'Product and service design (CSF13)' are the prominent influenced group factors. Analysis outcomes suggest that 'Top management commitment' is at the core of the efforts to maintain quality culture throughout the process. Furthermore, implication to theory and practices has been discussed, implication to theory will significantly contribute to existing literature and implication to practice will work as a guideline for practicing managers and industrial experts. At last, managers and quality engineers will get insight of how to develop efficient quality culture throughout organization by successful implementation of TQM.

Keywords Total Quality Management (TQM) Implementation · DEMATEL (Decision making Trial and Evaluation Method) · Grey set theory · Critical Success Factor (CSF's) · Grey based DEMATEL Method (Grey-DEMATEL) · Small and medium-sized enterprises (SMEs)

1 Introduction

We are moving towards a continuous varying global environment, customer's mindset, work culture & policies. These dynamics have pushed the organizations globally

to maintain and adopt new quality standards and goal that leads the organization to formulate new techniques, approaches, and strategies in their working environment to achieve high-quality standards for competitiveness. From management literature it has been observed that in order to improve organization's competitiveness and performance it is quite essential to formulate strategies on quality. According to Bhat and Rajashekhar (2009), there are three key approaches to achieve high-quality goals in organizations that are TQM, ISO-9000, and six-sigma. The linkage between environmental awareness, green practices, firm reputation and performance has been explained by Khan et al. (2021c).

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In past few decades, TQM has become one of the key approaches that have got excessive attention from both practitioners and academicians. Throughout the world TQM is seen as a technique that assists an organization to accomplish high-quality services and product (Ooi et al. 2011). Green Supply chain management, TQM and sustainable practices impact the performance of supply chain (Yu and Khan 2022; Khan et al. 2021a). It satisfies both customer's needs and expectations for the product in terms of quality and it also promotes organization competitiveness.

1.1 What is TQM?

In the total quality management (TQM) field of study Juran, Deming, Crosby, Ishikawa, and Feigenbaum are commonly known as Quality Gurus. By maintaining the same essence and spirit they defined TQM concept in their ways as:

- According to Juran's philosophy on TQM he defined quality as "fitness for use" and stated in the organization every person must be involved in the effort to make products or services that are fit for use.
- Similarly to improve productivity and performance of the organization Deming identified fourteen principles of quality management. Deming defined TQM as a continuous quality improvement process towards predictable degree of uniformity and dependability.
- According to Crosby point of view quality was defined as "conformance to requirements". Crosby mainly emphasized on zero defects and doing it right the first time.
- To improve organizational performance, Ishikawa mainly emphasized on the importance of total quality control. According to Ishikawa "quality not only means the quality of product, but also after sales service, quality of management, the company itself and the human life.

Based on different available philosophy by different quality gurus on TQM, it can be simply stated that it's basically a 'coherent' program or system of activities and policies which combines people, process & technology in an institutional infrastructure that provides vision incentive, support to the organization for achieving the goal of total quality.

The origin of TQM took place in the late 1950s in Japanese industries, and by 1980's it became more acceptable in western industries. Implementation of TQM in Japanese and western countries has resulted in significant growth and implementation that enhances the overall firm's performance in terms of organizational excellence (Goetsch and Davis 2013). Obstacles that hinder the adoption of social sustainability in the multi-tier supply chain has been explained by Khan et al. 2021a, b, d.

Organizations regularly maintain conducive environment of TQM to enhance and maintain customer satisfaction,

quality of product, productivity, teamwork, global competitiveness, market share, and profitability (Kumar et al. 2011).

Policy measures should boost technological endowment to improve their resilience in uncertainty and risk scenarios. Technological Innovation and Circular Economy Practices also plays an important role in improving the resilience in uncertainty and risk scenarios (Khan et al. 2021a).

The organizations in India are going through an environment that is transforming expeditiously in the face of globalization (Khanna et al. 2010). Globally quality has been observed as a more dominant strategic weapon (Kumar et al. 2009). To be in such a global competitive environment, Indian industries have realized that if didn't become more quality conscious they will not be able to succeed and attain competitive advantage. Although having known the strategic values and benefits of TQM, many Indian SMEs are still facing the issues of successful adoption of TQM.

According to small and medium business chamber of India;

"SMEs play a vital role in the growth of Indian economy by contributing 45% of the industrial output, 40% of exports, 42 million in employment, create one million jobs every year and produces more than 8000 quality products for the Indian and international markets."

According to Demirbag et al. 2006, SMEs success has a direct impact on both developed as well as developing countries economic development. SMEs are considered as the backbone of Indian economy (Sharma et al. 2014; Singh et al. 2006). However, evidence suggests that many SMEs have been unsuccessful in implementing TQM successfully (Ahire and O'Shaughnessy 1998). SMEs in India are quite slow in terms of TQM adoption as compared to large industry, as they consider that quality management practices in organization is of expensive and daunting prospect, having associated high on-going operational costs. (Sahoo and Yadav 2018) "One of the major reasons for the dismal performance in the world markets is the lack of effective quality management systems" (Khanna et al. 2002). Thus it becomes reasonably essential for Indian SMEs to adopt and implement TQM culture in their organization. To implement policies and quality culture in an organization, it's relevant to first identify critical success factors and then to find interrelationship among factors for successful TQM implementation.

From last decade, it has been observed several journals are addressing numerous quality management issues in organization, this enabled TQM as an area of academic interest among scholars in India (Kakkar and Narag 2007; Khanna 2009; Kumar et al. 2009; Bhat and Rajashekhar 2009; Prasad and Tata 2009).

According to Ismail Salaheldin 2009, in any setting, the preliminary step required prior to successful TQM implementation is to identify and prioritize its vital key enablers

(CSFs). A large number of studies that have put importance on identifying and prioritizing TQM enablers, generally denoted as (CSF's) (Saraph et al. 1989; Wali et al. 2003; Singh and Sushil 2013). Whereas failure in identification and prioritizing of CSFs before TQM implementation may significantly result in unsuccessful implementation of TQM and produce unacceptable results. (Jones and Seraphim 2008). There is still scarcity of studies on CSFs of TQM in India SMEs.

The present research work predominantly addresses the following two mainly quality-related objectives in Indian SMEs:

- To identify the Critical Success Factors for effective adoption of TQM implementation initiatives in the Indian SMEs;
- To uncover the causal relationships among the shortlisted Critical Success Factors and to categorize them into cause and effect groups for effective adoption of TQM implementation initiative in the Indian SMEs.

In this research work the key objective is to uncover the significant CSFs for successful TQM implementation in Indian SME's. From the system viewpoint, in this study, Grey based DEMATEL method has been applied to understand the structure of complicated causal relationships among identified 15 TQM implementation CSFs and classified them into cause (influential) and effect (resulting) groups. In the applied integrated method, decision making DEMATEL is a structural model (Ali et al. 2019) that has been applied to find causal interactions or to solve issue of dependencies (Hsu et al. 2016) among the elements with the help of graphs (Gandhi et al. 2016). In addition, Grey set theory is an effective approach to deal with discrete data having uncertainty (Deng 1982; Bai and Sarkis 2013), ambiguity and uncertainty into the analysis process (Luthra et al. 2017).

Also, the applied combined Grey set theory mathematics and Decision making Trial and Evaluation Method (Grey-DEMATEL) is able to effectively manage uncertain judgments (Rajesh and Ravi 2017) as well as flexibly deals with vagueness (Ali et al. 2019) and to evaluate cause and effect relations among factors (Bai and Sarkis 2013; Luthra et al. 2017). By combining the grey-based DEMATEL approach, the interrelation among the identified critical success factor is discussed for successfully TQM implementation in Indian SMEs.

The rest part of this paper is structured as follows: A review of the literature related to this study is provided in Section 2. CSFs to successful TQM implementation and data collection stages are illustrated in Section 3. The framework proposed in this research is presented in Section 4. The research methodology is explained in Section 5. Discussion on the research work findings and managerial implications are provided in Sections 6 and 7. Finally, conclusion,

limitations of the study and possibilities of future work are presented in Section 8.

2 Literature review

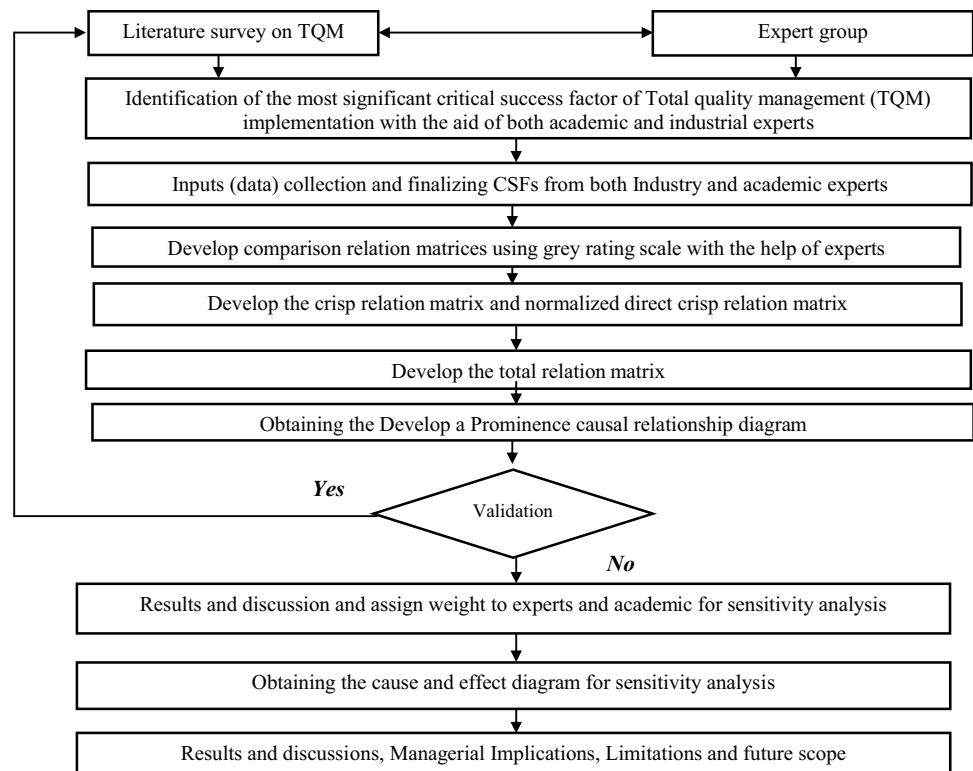
In the present research study, Fig. 1 illustrates the proposed research methodology. The applied integrated Grey based DEMATEL method is applied to uncover the causal relationships by diagraphs or matrices (Bai and Sarkis 2013). Among the identified CSF's under applied joint methodology had ability to visualize the structure of complicated causal relationships (Chang et al. 2011; Luthra et al. 2019) between CSFs and obtain the influence level of these factors (Rajesh and Ravi 2017).

2.1 Identification of Critical Success Factor (CSFs) for implementing Total Quality Management (TQM)

According to Oakland and Leslie (1996), term CSF is explained as "A term used to mean the most important sub-goals of a business organization". CSF's represent all the activities; those are quite necessary to accomplish the objectives of organization. They explained the CSF as what the organizations must implement in their working to achieve the organization's mission. This can be achieved by examination and categorization of the impact factors. These correspond to stable and higher-level processes that must be operating efficiently and effectively in order to produce the outputs and measurements, which constitute the objectives. In this contextual, it was felt utmost importance in Indian context to study CSFs (Wali et al. 2003).

A brief literature review in the field of TQM results in several CSFs depending on organization, type, size, internal/external factors is well illustrated in Table 1. The Significant literature on TQM is derived from the reviews inside the industrialized world (Gosen et al. 2005). A large number of studies have been conducted in several countries on TQM implementation. First in developed nations (e.g. Canada, Japan UK, and Australia) and later in developing nations like (e.g. Saudi Arabia, Ghana, Malaysia, India, China, and Taiwan).

Saraph et al. (1989) in his empirical study, in USA identified eight CSFs with sixty six elements from the concepts and perception of quality guru in the quality management field like role of divisional top management, quality policy, employee relations, quality data, reporting process management, supplier quality management, product or service design, training, and role of the quality department for TQM implementation. Similarly Black and Porter (1996) identified ten CSF factors of TQM with thirty two elements in the member-ship organization of European foundation for quality management like corporate quality culture, quality improvement measurement system, operational

Fig. 1 Proposed research methodology

quality planning, team work structure for improvement, strategic quality management, external interface management, customer satisfaction orientation, communication of improvement information, supplier partnership, and people & customer management.

However Khanna et al. (2011) study on Indian manufacturing industry identified ten CSFs for TQM implementation like top management commitment, suppliers quality management, human resources management, process management, customer focus, role of quality department, product design, quality information system and use of IT, training and quality citizenship.

A number of factors that affect the implementation of TQM may vary from country to country and industry to industry, more or less features may be different among countries, management of similar type doesn't guarantee to provide similar success of the actual implementation (Rezazadeh et al. 2012). Moreover in this study total fifteen CSF are identified w.r.t Indian SMEs with appropriate industrial expert's consultation.

2.2 Modelling approaches used for successful TQM implementation initiatives

A number of literatures are available in the context of identification of CSFs for TQM implementation in SMEs all over the globe. In which researchers applied various modeling tools, methodologies/techniques and developed framework for successful TQM implementation. A brief review of

modelling techniques used in TQM implementation initiatives are presented in Table 2.

In Table 2, modeling techniques such as BWM, AHP, ANP, DEMATEL, and ISM with TISM used by various researchers are presented. They have been utilized to address common issues of TQM implementation. However, no application of integrated grey-DEMATEL method is available in the context of TQM implementation. Thus the attempt of present research study is significant in the given domain. In addition there are number of studies which tried to identify interrelation between TQM- CSF like Khanam et al. (2015); Singh and Sushil (2013) applied integrated ISM with MICMAC and ISM with TISM method respectively to find interrelation among identified TQM enablers whereas Talib et al. (2011) by using ISM (Interpretive structural modeling) has been used to identify interrelationship among barrier for TQM implementation. Number of research work are dedicated to identify and rank the model using AHP, DEMATEL, Fuzzy-DEMATEL and BWM decision making method (Jamali et al. 2010; Rezazadeh et al. 2012; Mehralian et al. 2016; Gupta et al. 2017; Gupta et al. 2019; Talib et al. 2019). As compared to DEA/ANP/AHP and ISM method, the applied Grey based DEMATEL method is quite immature due to its limited applicability results in superior outcomes (Bai and Sarkis 2013). Applied methodology has also ability to tackle uncertainty issues in the cases having discrete data and incomplete information (Julong 1989; Ali et al. 2019). Various other modelling methodologies are not

Table 1 List of researcher with common CSFs with our study

S. No.	Researchers	Year	TMC	SQM	HRM	PM	CF&S	T&E	SD
1	Juran	1974	✓	✓		✓		✓	
2	Crosby	1979	✓					✓	
3	Deming	1986	✓					✓	
4	Saraph et al.	1989	✓	✓		✓	✓	✓	
5	Black and Porter	1996	✓				✓	✓	
6	Motwani	2001a, b	✓			✓			
7	Antony et al.	2002					✓		
8	Khoo and Tan	2002	✓				✓		
9	Chin et al.	2002	✓		✓			✓	
10	Bayazit	2003	✓				✓	✓	
11	Lewis et al.	2006		✓	✓			✓	
12	Das et al.	2008	✓	✓		✓		✓	
13	Prajogo	2008	✓			✓	✓		
14	Valmohammadi	2010			✓			✓	
15	Fotopoulos and Psomas	2010	✓				✓		
16	Khanna et al.	2011		✓	✓	✓	✓	✓	
17	Arumugam et al.	2011	✓	✓	✓	✓	✓	✓	
18	Gherbal et al.	2012		✓				✓	
19	Kutlu and Kadaifci	2014	✓			✓	✓	✓	
20	Manhas et al.	2015	✓	✓		✓	✓	✓	
21	Zhong et al.	2016	✓			✓	✓	✓	✓
22	Aquilani et al.	2017	✓		✓	✓	✓	✓	
23	Gupta et al.	2017		✓	✓			✓	
24	Yan et al.	2019	✓	✓	✓		✓		
25	Kaur et al.	2019	✓			✓	✓	✓	
S. No.	EI	TW	OC	IS	CI	P&S	IO	EE	
1		✓							
2									
3									
4	✓			✓		✓			
5		✓							
6					✓	✓			✓
7					✓				
8		✓							✓
9	✓		✓						
10	✓	✓							
11					✓				
12	✓				✓				
13	✓			✓					
14			✓	✓					
15	✓								
16				✓		✓			
17		✓				✓			
18		✓	✓	✓					
19	✓	✓		✓		✓			✓
20	✓	✓			✓				
21	✓								✓
22	✓			✓		✓			
23			✓				✓		

Table 1 (continued)

S. No.	EI	TW	OC	IS	CI	P&S	IO	EE
24			✓		✓			
25	✓	✓		✓	✓	✓		✓

well proficient as applied integrated Grey based DEMATEL to establish the strength of causal relations among CSFs.

In addition, the applied methodology has been extensively gaining its importance among researchers due to its ability to capture the interrelations among criteria (Ali et al. 2019). A brief review of the applied Grey based DEMATEL methodology in different studies is illustrated in Table 3.

According to Table 3, numerous researchers have applied integrated Grey-DEMTEL method to address various issues like Bai and Sarkis (2013), applied in his business process study to evaluate business process management (BPM) CSFs. Whereas Ali et al. 2019 applied it in the food supply chain study to evaluate risks. However Gupta and Barua (2018), models the enablers of green innovation and Shao et al. (2014) used it to identify the barriers and relationship between green products and consumers. Similarly Rajesh and Ravi (2015), used it in the field of supply chain to model enablers of supply chain risk mitigation.

2.3 Research gaps and highlights

Overall objective of implementing TQM philosophy in industries is to continuously improve the product quality to attain customer satisfaction (Joseph et al. 1999). Today the era of high worldwide business competition pushed the

organization for substantial change in the business process. Khanna et al. 2011 investigated several CSF's of TQM and prioritized them by applying decision-making technique for Indian manufacturing industry. Kr Singh (2011) identified TQM implementation CSFs and tried to develop a structural relationship among factors under study by using Interpretive structural modelling (ISM) method. Arumugam et al. (2011) identified several CSFs and tried to find their impact on performance in industry. Lau and Idris (2001) used multiple regression analysis and binomial testing in their data collected through questionnaire survey and by analyzing they structured and developed the model for TQM implementation in industry.

It can be noted that a research gap exists in investigating critical success factor and analyzing the interactions among identified CSF's. This study work can be utilized as a guideline for practice managers and executives that will guide them while formulating total quality management strategies for their respective industries. Furthermore, proposed framework can be used for any industry having more or fewer TQM implementation factors. In this paper, Grey-based DEMATEL framework has been developed for categorizing and for the purpose to evaluate the contextual relationships among identified TQM CSFs in automobile industry operating in northern part of India.

Table 2 Several modelling techniques/methodologies to address issues in TQM

S. No.	Researcher (Year)	Methods/Approaches applied	Issues addressed
1	Karuppusami and Gandhinathan (2006)	Pareto analysis	Pareto analysis of critical success factors of TQM
2	Bayazit and Karpak (2007)	Analytical network process (ANP)	Framework for successful TQM implementation
3	Jamali et al. (2010)	DEMATEL	Identifying and analyzing the TQM CSFs
4	Talib et al. (2011)	Interpretive structural modeling (ISM)	Analysis of interaction among the barriers of TQM
5	Rezazadeh et al. (2012)	Fuzzy-AHP (Analytical hierarchy process)	Evaluating and prioritizing CSF of TQM implementation
6	Singh and Sushil (2013)	Interpretive structural modeling (ISM) and Total Interpretive structural modeling (TISM)	Identifying and analyzing the interactions among different enablers of TQM
7	Khanam et al. (2015)	ISM-MICMAC approach	Modeling the TQM enablers
8	Mehralian et al. (2016)	TOPSIS (Technique for order of preference by similarity to ideal solution)	Prioritizing CSFs of TQM
9	Gupta et al. (2017)	Fuzzy-DEMATEL approach	Evaluating TQM adoption success factors
10	Jusoh et al. (2018)	Fuzzy Delphi with Fuzzy AHP	Evaluating CSFs of TQM
11	Anil and K.P. (2019); Sohal and Terziovski (2000)	Structural equation modelling (SEM)	TQM practices and its performance effects
12	Dilawo and Salimi (2019)	Quality function Deployment (QFD)	Understanding TQM implementation barriers
13	Talib et al. (2019)	Best–Worst Method (BWM)	Ranking model of TQM enablers
14	Pambreni et al. (2019)	Multiple linear regression analysis	Influence of TQM towards organization performance

Table 3 Issues addressed using Grey-based DEMATEL method in various literatures

S. No.	Researcher (Year)	Issue's addressed using Grey-DEMATEL method
1	Ying et al. (2013)	To analyze barriers for Green Food Sustainable Supply Chain
2	Bai and Sarkis (2013)	To evaluate business process management (BPM) critical success factors
3	Shao et al. (2014)	To identify barriers and relationship between green products and consumers
4	Xia et al. (2015)	To analyze internal barriers for SMES parts remanufacturers
5	Rajesh and Ravi (2015)	To model enablers of supply chain risk mitigation
6	Liang et al. (2016)	To identify CSFs for sustainable development
7	Ozcan and Tuysuz (2016)	For evaluating performance of retail stores
8	Su et al. (2016)	To improve sustainable supply chain management
9	Luthra et al. (2017)	Performed structural modelling for sustainable consumption and production adoption
10	Bhatia and Srivastava (2018)	For analysis of external barriers to remanufacturing
11	Luthra et al. (2018)	For analyzing the barriers between ecofriendly products and consumers
12	Gupta and Barua (2018)	To model enablers of green innovation
13	Ali et al. (2019)	To evaluate risks in the food supply chain
14	Haleem et al. (2019)	For traceability implementation in the food supply chain
15	Yazdani et al. (2021)	Developed framework for agriculture supply chain risk management under a circular economy context

This research contribution includes:

- Finding net-prominence among the identified CSF's and categorizing them into cause and effect group.
- Identifying various CSF's based on a literature review and with the response of industrial experts' for the case industry and evaluating the contextual relations by utilizing a grey-based DEMATEL framework.
- Investigating of cause and effect relations among the identified CSF's used in the GREY-DEMATEL framework.
- Proposing implications to both theory and practice point of view. Implication to practice will work as a guideline for practice managers and industrial experts. Whereas implication proposed to theory will significantly contribute to the existing literature.

3 Data collection stages and definitions of TQM CSFs

3.1 Data collection

For the data collection process, three experts from industries and one expert from academics having immense experience in the field of TQM were consulted. The data collection process was completed in number of phase as described below:

3.1.1 Experts and CSF'S identification

Experts were identified from the industries where TQM was already implemented successfully for more

than 10 years. These experts were of the level of TQM head and General Manager, quality control. Additionally, one expert from academic field with vast teaching experience in the field of statistical quality control was also chosen to give its opinion on the subject matter. Data collection regarding task was accomplished in two stages, as defined below:

Stage 1: Finalizing the relevant TQM CSFs for Indian Industries Initially, by reviewing various research journals, TQM CSF's were identified in various organizations. A total of 20 CSF's were identified and discussed with experts. Based on their industry, experts were requested to delete or add TQM CSF, finally they shortlisted them into 15 most relevant TQM CSF from the perspective of Indian SME's. The collection of their response was accomplished by set of questioner and discussion session was arranged for information consolidation.

Available expert input was utilized to identify the most relevant factors. To accomplish the objective, responses from the experts were used to prioritize the shortlisted CSF's. For prioritizing the factors a triangular fuzzy membership function based DEMATEL model was developed.

Stage 2: Evaluation of the comparison of identified TQM CSF's The research methodology and objectives of the research were discussed with expert panel and they were requested to give their response in a pairwise comparison matrix, that was essentially required for developing a triangular fuzzy membership function based DEMATEL model.

The considered TQM CSF's in this study and associated literature are presented in Table 1. Additionally, the summary of the code applied to identify the utmost factor is presented in Table 8.

3.2 Definitions

3.2.1 Top management commitment

In organizations it seems as an initial step toward successful TQM implementation through leadership strategy and policies, managing innovation, management by fact and quality planning. Top management commitment in a manufacturing plant can maintain customer focus, employee empowerment, and management of quality of the supplier. In organization it measures the involvement and support for quality of individuals. Senior managers' involvement, understanding and customer focus are essential antecedents of TQM success (Taylor and Wright 2003; Kumar et al. 2011).

3.2.2 Supplier quality management

A supplier having good product quality, technical and financial capabilities, higher efficiency in operation and on-time delivery results in increases of overall product quality. So, effective supplier quality management offers a high-quality product and is enviable. An enduring relationship, as well as good coordination among supplier and manufacturers, is highly desirable. The quality of the end product is directly related to the quality of raw materials and equipment supplied by the vendors (Khanna et al. 2011).

3.2.3 Human resource management

The main focus of human resource management is on employee empowerment to make them efficient to contribute effectively to the quality processes. It emphasizes on having an open culture, teamwork and continuous improvement of employees in an organization. It helps to attain customer requirements by providing good quality products. According to Khanna et al. 2011, employees also need to be aware of their roles and goals and are trained for their jobs.

3.2.4 Process management

TQM emphasizes on continuous improvement of processes involved in an organization. Effective process management includes a reduction in cycle time for order processing, improvement of product and service and on-time delivery of the product. It stresses on plans to continuously reduce

defects, continuous value addition in the process, increasing each employee productivity and quality improvement of the organization (Motwani 2001a, b; Talib 2013).

3.2.5 Customer focus and satisfaction

The effectiveness of an organization is proportional to customer's satisfaction. For long term profitability, organization must focus on the expectation of customers and work according to the customer's expectations which will overall improve the quality of the product. (Talib and Rahman 2010). Customer focus also comprises of effective relationships with customer. It's a central tenet of market orientation; it is a set of beliefs that puts the customer's interest first (Manhas et al. 2015). It's needed for each organization to systematically capture requirements of customers and ensure level of satisfaction after sales (Khanna et al. 2011).

3.2.6 Training and education

Training and education of employees in an organization increases the efficiency of employees towards their work. Proper training benefits employees to manage methodology of total quality management. It prepares employees to adopt new technologies and skills for the development of the product. Necessary education makes employees get aware of the new evolving methodology and processes.

3.2.7 Supplier development

There is wide quality commitment associated in supply chain to attain excellence in production and service management from the supplier to the producer to the consumer-of an organization. In TQM implementation success, this factor holds equal importance. (Christofi et al. 2008; Gupta et al. 2017). In modern business there is interdependence of buyers and supplier (Kr Singh 2011). Several quality problems have their source in defective incoming supplies. To gain competitive advantage in global trade and business the relationship with suppliers plays a major role to attain TQM success (Flynn et al. 1994).

3.2.8 Employee involvement

Employee involvement emphasizes the contribution of employees in decision making and problem-solving processes at all levels and active participation for new ideas and innovations. According to Oliver 1998, involvement means to empower employees, give them information, increase their knowledge and reward quality performance (Talib and Rahman 2010). It encourages employee's optimistic approach and behavior in the organization. It helps in continuous improvement at every level of an organization. Direct active employee involvement to the vision, values

and quality goals are necessary for success of TQM program (Motwani 2001a, b; Singh and Sushil 2013).

3.2.9 Team work

Teamwork focuses on the commitment of the employees to work together for the organization's goals with good coordination. Involvement of employees at all levels, continuous learning and flexibility are the results of teamwork in an organization and increases productivity. It's a central to managing change and implementing plans, solving problems and creating empathy and a sense of involvement (Talib and Rahman 2010). Teamwork results in more committed and involved employees within the organization (Manhas et al. 2015).

3.2.10 Existing organizational culture

A good organizational culture is directly proportional to the effectiveness and performance of the organization. Healthy organizational culture encourages employees for continuous improvement and development of the organization. Organizational culture with flexible structure and an encouraging climate are more beneficial to the successful implementation of total quality management in an organization.

3.2.11 Information system

Implementation of information systems and technology in TQM can develop operational tasks of quality management and hence improve the quality of the product. Application of information system in TQM focuses on database management, data processing, replication, inventory control, warehouse management, network and effective communication in an organization.

3.2.12 Continuous improvement

Aim of continuous improvement in an organization is to obtain constant progress and developing processes for innovative methods in the process of development of output. Continuous improvement of the process helps in the reduction of the process variability, hence improving the output. It helps the organization to survive in the competitive market (Antony et al. 2002). According to Turney and Anderson (1989), continuous improvement act as the relentless pursuit of improvement in the delivery of value to customers.

3.2.13 Product and service design

Product and service design comprises of a generation of new ideas, concept development, development of new

methodologies, implementation of innovative technologies and processes. Product and service design focuses on design of the specification of the product according to the needs of the customers, which increases productivity and market value of the organization. (Motwani (2001a, b); Jamali et al. (2010); Talib and Rahman (2010)). Product and service design relates to the products clarity and process specifications, the totality of new product tests and inspections before entering the market (Kr Singh 2011).

3.2.14 Inter-organizational linkages

Strategic collaboration among organizations and industries helps to maintain quality in the organization and to sustain in the competitive market (Gupta et al. 2017). Forming an alliance with organization helps to share the risks and capabilities, which is beneficial for quality management.

3.2.15 Employee empowerment

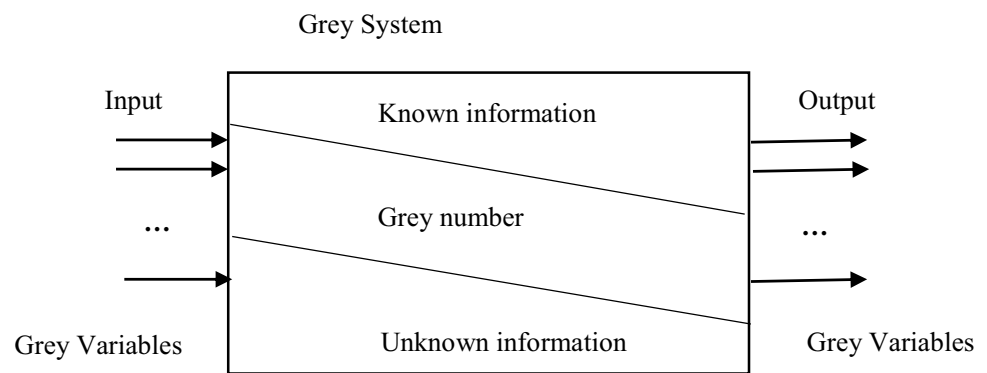
Employee empowerment focuses on the involvement of all employees in the decision making and problem-solving processes and their contribution to the work development of the organization. By providing proper training and education and equal importance, employees of an organization can be empowered to work towards the goal of the organization. Employee empowerment is a key aspect of the implementation of total quality management in an organization (Thiagaragan et al. 2001; Singh and Sushil 2013).

4 Integrated grey system theory and DEMATEL Methodology for evaluation of causal and effective factor relationships

4.1 Introducing the grey system theory

The ability of the grey set theory is to tackle uncertainty issues in the cases having discrete data and incomplete information. It can be seen as a powerful methodology that produces satisfactory results at conditions where variability in factors is present. Deng (1982) proposed grey theory methodology which was discovered from grey set. In past recent studies this methodology results as an effective approach to deal with discrete data consist of uncertainty. The key cause of imprecision comprises of partial ignorance, non-obtainable information, data with incomplete information, unquantifiable information (Bai and Sarkis 2013; Deng 1982, Li et al. 1997).

Fig. 2 Concept of grey system.
Source: (Tseng 2009)



4.2 Mathematical operations associated with grey numbers

Let 'a' signifies closed and bounded set of real numbers.

Let $\otimes a_1$ and $\otimes a_2$ are two grey number, the mathematical operations associated with these no.'s are as follows:

1. Addition of two grey numbers:

$$\otimes a_1 + \otimes a_2 = \left[\begin{matrix} a_1 + a_2, \bar{a}_1 + \bar{a}_2 \\ - \end{matrix} \right]$$

2. Subtraction of two grey numbers:

$$\otimes a_1 - \otimes a_2 = \left[\begin{matrix} a_1 - \bar{a}_2, \bar{a}_1 + a_2 \\ - \end{matrix} \right]$$

3. Multiplication of two grey numbers:

$$\otimes a_1 \times \otimes a_2 = \left[\begin{matrix} \min \left(\begin{matrix} a_1 \cdot a_2, a_1 \cdot \bar{a}_2, \bar{a}_1 \cdot a_2, \bar{a}_1 \cdot \bar{a}_2 \\ - \end{matrix} \right), \\ \max \left(\begin{matrix} a_1 \cdot a_2, a_1 \cdot \bar{a}_2, \bar{a}_1 \cdot a_2, \bar{a}_1 \cdot \bar{a}_2 \\ - \end{matrix} \right) \end{matrix} \right]$$

4. Division of two grey numbers:

$$\otimes a_1 \div \otimes a_2 = \left[\begin{matrix} a_1, \bar{a}_1 \\ - \end{matrix} \right] \times \left[\begin{matrix} 1/\bar{a}_2, 1/a_2 \\ - \end{matrix} \right]$$

4.3 Introducing DEMATEL methodology

The DEMATEL MCDM method based on graph theory was introduced by the Battelle Memorial Institute. It's

widely applied MCDM methodology used to build and analyze the structural model of complicated causal relationship amongst a set of factors by diagrams or matrices. This MCDM method is not just limited to transform inter-dependency relationships into cause & effect groups by metrics but it also discovers the critical factors by the aid of impact-relation diagram in complex structure system (Figs. 2 and 3).

In the DEMATEL method diagrams or matrices portrays the contextual relationship between the set of elements of the system. This method converts the contextual relation amongst the causes & effects into structural model.

5 Methodology

The basic steps in the applied grey-based DEMATEL Methodology (Bai and Sarkis (2013); Luthra et al. (2017)) are as follows:

1. For each decision-maker inputs, form the fuzzy-based direct relation matrix:

Based on literature analysis and investigation the influencing factor and their relationship are recognized. By defining the Grey Linguistic scale presented in Table 4, scale holding the scale items factor influence relationships in terms of five linguistic terms. Here, for each linguistic term there is its corresponding grey number.

The influence relationship measure between the CSF's ($\alpha = \alpha_1, \alpha_2, \dots, \alpha_n$) can be measured as:

The 'X' no. of decision-makers of the same field, asked to give their inputs/response based on following five linguistic terms like (NI- [No influence = 0], VLI-[very low

Fig. 3 Procedure in DEMATEL methodology

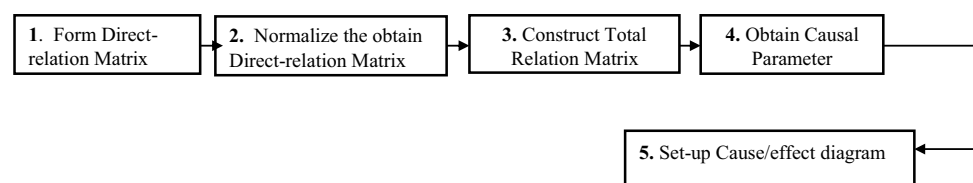


Table 4 The respondent's assessments of linguistic terms and grey number

Linguistic term	Grey numbers
No influence (N)	[0,0]
Very low influence (VL)	[0,1]
Low influence (L)	[1,2]
High influence (H)	[2,3]
Very high influence (VH)	[3,4]

Source: (Bai and Sarkis 2013)

influence = 1], LI-[low influence = 2], HI-[High influence = 3], VHI-[Very high influence = 4]) in the pairwise comparison ($n \times n$) matrix by analyzing the direct impact of individual factor on other considered factor. For each decision-maker there is corresponding grey metrics (X_1 ; X_2 ;; X_n) and its elements analogous to grey number. Obtained grey-based direct relation matrix ($\mathcal{M}^k$) can be represented as in Eq. (1).

$$\mathcal{M}^x = \begin{matrix} \alpha 1 \\ \alpha 2 \\ \vdots \\ \alpha n \end{matrix} \begin{bmatrix} [0,0] & \otimes \mathcal{M}_{12}^x & \cdots & \otimes \mathcal{M}_{1n}^x \\ \otimes \mathcal{M}_{21}^x & [0,0] & \cdots & \otimes \mathcal{M}_{2n}^x \\ \vdots & \vdots & \ddots & \vdots \\ \otimes \mathcal{M}_{n1}^x & \otimes \mathcal{M}_{n2}^x & \cdots & [0,0] \end{bmatrix} \quad (1)$$

In expression (1) [$\otimes \mathcal{M}^x$] denotes a grey number for the influence of TQM implementation CSF's i on the j .

$$\begin{pmatrix} \mathcal{T} \end{pmatrix} = \mathcal{M}_{Normalized} + \mathcal{M}_{Normalized}^2 + \mathcal{M}_{Normalized}^3 + \dots = \sum_{i=1}^{\omega} \mathcal{M}_{Normalized}^i \quad (6)$$

$= [\mathcal{M}_{Normalized} \times (\mathbf{I} - \mathcal{M}_{Normalized})^{-1}]$; where $\mathbf{I}$ = identity matrix of size $n \times n$.

2. Form aggregated matrix $\mathcal{M}$

By combining all individual grey-based direct relation matrices ($\mathcal{M}^k$), via the process of averaging, is given by expression (2).

$$\mathcal{M} = (\sum_{i=1}^x \mathcal{M}^x) / x \quad (2)$$

The procedure for aggregating is completed for every expert direct relation matrix. Table 5 represents the aggregated direct relation matrix ($\mathcal{M}$).

3. Obtain Normalized grey direct relation matrix ($\mathcal{M}_{Normalized}$):

The obtained aggregated grey direct relation matrix ($\mathcal{M}$) can be transformed into the normalized grey direct relation matrix ($\mathcal{M}_{Normalized}$) by using Expressions (3) (4), for the presented case study Table 6 presents the ($\mathcal{M}_{Normalized}$) matrix.

$$\otimes a = [\underline{a}, \bar{a}] = \frac{1}{\sum_{j=1}^n \otimes \mathcal{M}_{ij}}; \text{ Where } i, j = \overline{1, n} \quad (3)$$

$$\mathcal{M}_{Normalized} = \mathcal{M}_N = \otimes a \cdot \mathcal{M} \quad (4)$$

$$\Downarrow_{ij} = \left[\frac{a \cdot \mathcal{M}_{ij}}{ij}, \frac{\bar{a} \cdot \mathcal{M}_{ij}}{ij} \right] \quad (5)$$

Form Table 5, the value of $\sum_{j=1}^n \otimes \mathcal{M}_{ij}$ for every, i can be calculated simply by adding i^{th} row values. For example in presented case study $\sum_{j=1}^n \otimes \mathcal{M}_{ij}$, values for $i=1, \dots, 15$ given as

[31.2,45.2],[24.2,36.6],[25.6,39],[26.2,40],[27.4,40.6],[27.4,41.4],[26.4,40],[25.8,39.8],[27.4,41],[27.2,41],[28.2,42.2],[25.8,38.8],[25.4,39],[25.39],[26.39.2]

In addition to this for all, i row in the given Table 5, the maximum value of $\sum_{j=1}^n \otimes \mathcal{M}_{ij}$, results in: $\otimes a = \left[\frac{1}{45.2}, \frac{1}{31.2} \right]$. By expression (4), the value of $\mathcal{M}_{Normalized} = \otimes a \cdot \mathcal{M}$ can be determined as $\left[\frac{a}{45.2}, \frac{\bar{a}}{31.2} \right]$, similarly for all rest other element values in the normalized form of grey relation matrix $\mathcal{M}_{Normalized}$ is presented in Table 6.

4. Obtain the total relation-matrix ($\mathcal{T}$):

By using expression (6) determine the total relation matrix ($\mathcal{T}$).

Table 7 of size 15×15 , represents each element value of $\mathcal{T}$.

5. For every i^{th} row and j^{th} column, by using expression (7), (8), calculate the sum of row ($\mathcal{R}_i$) and column ($\mathcal{D}_j$) in the obtained total relation-matrix ($\mathcal{T}$).

$$\otimes \widetilde{\mathcal{R}}_i = \sum_{j=1}^n t_{ij} \forall i; \quad (7)$$

$$\otimes \widetilde{\mathcal{D}}_j = \sum_{i=1}^n t_{ij} \forall j; \quad (8)$$

Where $\otimes \widetilde{\mathcal{R}}_i$ denotes the row sum values direct and indirect influence by TQM implementation CSF i on the other CSF. In addition to this $\otimes \widetilde{\mathcal{D}}_j$ denotes the column sum of direct and in-direct influence by TQM implementation CSF j is receiving from the other CSF's.

6. Calculate the Prominence ($\otimes \widetilde{\mathcal{P}}_i$) and net effect ($\otimes \widetilde{NE}_i$):

Table 5 The overall fuzzy direct relation matrix

	CSF1	CSF2	CSF3	CSF4	CSF5	CSF6	CSF7	CSF8
CSF1	[0,0]	[2.2,3.2]	[2.2,3.2]	[2,3]	[2.8,3.8]	[2,3]	[2.2,3.2]	[2.2,3.2]
CSF2	[1.8,2.6]	[0,0]	[0.8,1.6]	[1.2,2]	[2.2,3.2]	[1.2,2]	[2,3]	[1.8,2.8]
CSF3	[1.6,2.4]	[1.2,2.2]	[0,0]	[1.8,2.8]	[1.6,2.6]	[2.6,3.6]	[1.4,2.4]	[1.8,2.8]
CSF4	[1.8,2.8]	[1.8,2.8]	[1.4,2.4]	[0,0]	[2.4,3.4]	[2,3]	[2.2,3.2]	[1.8,2.8]
CSF5	[2.4,3.4]	[2.2,3.2]	[1.2,2.2]	[2.4,3.4]	[0,0]	[1.8,2.8]	[2.2,3.2]	[1.6,2.6]
CSF6	[1.2,2.2]	[1.6,2.6]	[2,3]	[2.6,3.6]	[2.2,3.2]	[0,0]	[1.6,2.6]	[2,3]
CSF7	[2,3]	[2.2,3.2]	[1,1.8]	[2.6,3.6]	[2.4,3.4]	[1.2,2.2]	[0,0]	[1.8,2.8]
CSF8	[1.4,2.4]	[1.4,2.4]	[1.8,2.8]	[2.2,3.2]	[2,3]	[1.2,2.2]	[2.2,3.2]	[0,0]
CSF9	[1.4,2.4]	[2,3]	[1.8,2.8]	[2.2,3.2]	[2.6,3.6]	[1.2,2.2]	[2.4,3.4]	[2.6,3.6]
CSF10	[1,2]	[1.8,2.8]	[1.8,2.8]	[2.4,3.4]	[2.2,3.2]	[1.6,2.6]	[2.6,3.6]	[2,3]
CSF11	[2,3]	[2.2,3.2]	[2,3]	[2.6,3.6]	[2.4,3.4]	[1.4,2.4]	[2.2,3.2]	[2,3]
CSF12	[1.4,2.4]	[1.6,2.6]	[1.4,2.2]	[2.2,3.2]	[2.4,3.4]	[1.6,2.4]	[2.4,3.4]	[2,3]
CSF13	[2,3]	[2.2,3.2]	[0.8,1.6]	[2.6,3.6]	[2.6,3.6]	[1.8,2.8]	[2.2,3.2]	[1.6,2.6]
CSF14	[2.4,3.4]	[1.8,2.8]	[1.4,2.4]	[2.2,3.2]	[2,3]	[1.4,2.4]	[2,3]	[1.8,2.8]
CSF15	[1.4,2.2]	[1.2,2]	[2,3]	[2.4,3.4]	[1.8,2.8]	[1.8,2.8]	[2.2,3.2]	[2.4,3.4]
	CSF9	CSF10	CSF11	CSF12	CSF13	CSF14	CSF15	
CSF1	[2.2,3.2]	[2.2,3.2]	[2.4,3.4]	[2,3]	[2.4,3.4]	[2.6,3.6]	[1.8,2.8]	
CSF2	[2.2,3]	[2.2,8]	[1.8,2.8]	[2.2,3]	[2.2,3.2]	[1.8,2.8]	[1,1.8]	
CSF3	[2.2,3.2]	[2.6,3.6]	[1.8,2.8]	[2,3]	[0.8,1.6]	[1.6,2.6]	[2.4,3.4]	
CSF4	[2,3]	[1.6,2.6]	[1.6,2.6]	[2.4,3.4]	[2.4,3.4]	[1.6,2.6]	[1.2,2]	
CSF5	[2,3]	[1.8,2.6]	[1.8,2.6]	[2.2,8]	[2.4,3.4]	[2,3]	[1.6,2.4]	
CSF6	[1.8,2.8]	[2.2,3.2]	[2.4,3.4]	[2.4,3.4]	[2.4,3.4]	[1.2,2.2]	[1.8,2.8]	
CSF7	[2.2,3.2]	[2.4,3.4]	[1.6,2.6]	[2,3]	[1.6,2.4]	[1.8,2.8]	[1.6,2.6]	
CSF8	[2.2,3.2]	[2.2,3.2]	[2,3]	[2,3]	[1.8,2.8]	[1.6,2.6]	[1.8,2.8]	
CSF9	[0,0]	[2.8,3.8]	[2,3]	[1.4,2.4]	[1.6,2.6]	[1.2,2.2]	[1.8,2.8]	
CSF10	[1.8,2.8]	[0,0]	[1.8,2.8]	[2.2,3.2]	[2,3]	[1.8,2.8]	[2,3]	
CSF11	[1.6,2.6]	[2,3]	[0,0]	[2,3]	[2.2,3.2]	[1.8,2.8]	[1.8,2.8]	
CSF12	[1.8,2.8]	[1.8,2.8]	[1.6,2.4]	[0,0]	[2.4,3.4]	[1.4,2.2]	[1.8,2.6]	
CSF13	[1.4,2.2]	[2,3]	[1.4,2.4]	[1.6,2.6]	[0,0]	[1.6,2.6]	[1.6,2.6]	
CSF14	[1.4,2.4]	[1.8,2.8]	[1.6,2.6]	[1.6,2.6]	[1.8,2.8]	[0,0]	[1.8,2.8]	
CSF15	[2,3]	[2,3]	[1.4,2.2]	[2,3]	[2,3]	[1.4,2.4]	[0,0]	

By using expressions (9) and (10) obtain the values of $(\otimes \widetilde{P}_i)$ and $(\otimes \widetilde{NE}_i)$ of TQM implementation CSF i

$$(\otimes \widetilde{P}_i) = [\otimes \widetilde{R}_i + \otimes \widetilde{D}_j] ; \quad (9)$$

$$(\otimes \widetilde{NE}_i) = [\otimes \widetilde{R}_i - \otimes \widetilde{D}_j] ; \quad (10)$$

7. Develop a Prominence causal relationship graph:

Based on the obtained overall values of $[\otimes \widetilde{R}_i + \otimes \widetilde{D}_j]$ & $[\otimes \widetilde{R}_i - \otimes \widetilde{D}_j]$ in Table 8, plot these values on the two-dimensional axis for each CSF of TQM. The graphical representation of the overall aggregated barriers prominence and net effect results is presented in Fig. 5.

The overall prominence causal relationship diagram represents the relationship and pattern among all the considered CSF simultaneously and in pairs. The interrelationship in between each TQM CSF's is presented in Fig. 5 that is represented by the arrow (diagraphs). In Fig. 5 the mapping of relationship in between TQM CSF is done only for those factors whose value is above the value of threshold (ϕ) in a matrix $\begin{pmatrix} T \end{pmatrix}$. The 'threshold value (ϕ) is a sum of standard deviation and mean'. For the presented study the value of (ϕ) = [0.05014, 0.4584]. In Matrix $\begin{pmatrix} T \end{pmatrix}$, the factors value higher than the threshold value are underlined and are presented Table 7. The diagram representing the dyadic relationship is presented in Fig. 5. In this graph, both one and two-way significant relationship are represented by solid and dashed line respectively.

Table 6 Normalized Direct relation mat

	CSF1	CSF2	CSF3	CSF4	CSF5	CSF6	CSF7	CSF8
C1	[0,0]	[0.071,0.071]	[0.071,0.071]	[0.064,0.066]	[0.09,0.084]	[0.064,0.066]	[0.071,0.071]	[0.071,0.071]
C2	[0.058,0.058]	[0,0]	[0.026,0.035]	[0.038,0.044]	[0.071,0.071]	[0.038,0.044]	[0.064,0.066]	[0.058,0.062]
C3	[0.051,0.053]	[0.038,0.049]	[0,0]	[0.058,0.062]	[0.051,0.058]	[0.083,0.08]	[0.045,0.053]	[0.058,0.062]
C4	[0.058,0.062]	[0.058,0.062]	[0.045,0.053]	[0,0]	[0.077,0.075]	[0.064,0.066]	[0.071,0.071]	[0.058,0.062]
C5	[0.077,0.075]	[0.071,0.071]	[0.038,0.049]	[0.077,0.075]	[0,0]	[0.058,0.062]	[0.071,0.071]	[0.051,0.058]
C6	[0.038,0.049]	[0.051,0.058]	[0.064,0.066]	[0.083,0.08]	[0.071,0.071]	[0,0]	[0.051,0.058]	[0.064,0.066]
C7	[0.064,0.066]	[0.071,0.071]	[0.032,0.04]	[0.083,0.08]	[0.077,0.075]	[0.038,0.049]	[0,0]	[0.058,0.062]
C8	[0.045,0.053]	[0.045,0.053]	[0.058,0.062]	[0.071,0.071]	[0.064,0.066]	[0.038,0.049]	[0.071,0.071]	[0,0]
C9	[0.045,0.053]	[0.064,0.066]	[0.058,0.062]	[0.071,0.071]	[0.083,0.08]	[0.038,0.049]	[0.077,0.075]	[0.083,0.08]
C10	[0.032,0.044]	[0.058,0.062]	[0.058,0.062]	[0.077,0.075]	[0.071,0.071]	[0.051,0.058]	[0.083,0.08]	[0.064,0.066]
C11	[0.064,0.066]	[0.071,0.071]	[0.064,0.066]	[0.083,0.08]	[0.077,0.075]	[0.045,0.053]	[0.071,0.071]	[0.064,0.066]
C12	[0.045,0.053]	[0.051,0.058]	[0.045,0.049]	[0.071,0.071]	[0.077,0.075]	[0.051,0.053]	[0.077,0.075]	[0.064,0.066]
C13	[0.064,0.066]	[0.071,0.071]	[0.026,0.035]	[0.083,0.08]	[0.083,0.08]	[0.058,0.062]	[0.071,0.071]	[0.051,0.058]
C14	[0.077,0.075]	[0.058,0.062]	[0.045,0.053]	[0.071,0.071]	[0.064,0.066]	[0.045,0.053]	[0.064,0.066]	[0.058,0.062]
C15	[0.045,0.049]	[0.038,0.044]	[0.064,0.066]	[0.077,0.075]	[0.058,0.062]	[0.058,0.062]	[0.071,0.071]	[0.077,0.075]
	CSF9	CSF10	CSF11	CSF12	CSF13	CSF14	CSF15	
C1	[0.071,0.071]	[0.071,0.071]	[0.077,0.075]	[0.064,0.066]	[0.077,0.075]	[0.083,0.08]	[0.058,0.062]	
C2	[0.071,0.066]	[0.064,0.062]	[0.058,0.062]	[0.071,0.066]	[0.071,0.071]	[0.058,0.062]	[0.032,0.04]	
C3	[0.071,0.071]	[0.083,0.08]	[0.058,0.062]	[0.064,0.066]	[0.026,0.035]	[0.051,0.058]	[0.077,0.075]	
C4	[0.064,0.066]	[0.051,0.058]	[0.051,0.058]	[0.077,0.075]	[0.077,0.075]	[0.051,0.058]	[0.038,0.044]	
C5	[0.064,0.066]	[0.058,0.058]	[0.058,0.058]	[0.064,0.062]	[0.077,0.075]	[0.064,0.066]	[0.051,0.053]	
C6	[0.058,0.062]	[0.071,0.071]	[0.077,0.075]	[0.077,0.075]	[0.077,0.075]	[0.038,0.049]	[0.058,0.062]	
C7	[0.071,0.071]	[0.077,0.075]	[0.051,0.058]	[0.064,0.066]	[0.051,0.053]	[0.058,0.062]	[0.051,0.058]	
C8	[0.071,0.071]	[0.071,0.071]	[0.064,0.066]	[0.064,0.066]	[0.058,0.062]	[0.051,0.058]	[0.058,0.062]	
C9	[0,0]	[0.09,0.084]	[0.064,0.066]	[0.045,0.053]	[0.051,0.058]	[0.038,0.049]	[0.058,0.062]	
C10	[0.058,0.062]	[0,0]	[0.058,0.062]	[0.071,0.071]	[0.064,0.066]	[0.058,0.062]	[0.064,0.066]	
C11	[0.051,0.058]	[0.064,0.066]	[0,0]	[0.064,0.066]	[0.071,0.071]	[0.058,0.062]	[0.058,0.062]	
C12	[0.058,0.062]	[0.058,0.062]	[0.051,0.053]	[0,0]	[0.077,0.075]	[0.045,0.049]	[0.058,0.058]	
C13	[0.045,0.049]	[0.064,0.066]	[0.045,0.053]	[0.051,0.058]	[0,0]	[0.051,0.058]	[0.051,0.058]	
C14	[0.045,0.053]	[0.058,0.062]	[0.051,0.058]	[0.051,0.058]	[0.058,0.062]	[0,0]	[0.058,0.062]	
C15	[0.064,0.066]	[0.064,0.066]	[0.045,0.049]	[0.064,0.066]	[0.064,0.066]	[0.045,0.053]	[0,0]	

6 Results and discussion

The factors having the most prominence and their relationships amongst the TQM implementation CSF's can be observed from Figs. 4 and 5, where customer focus (CSF5), process management (CSF4), supplier development (CSF7), existing organizational culture (CSF10) and top management commitment (CSF51) were the five topmost prominence factors. Results obtained from various study revealed that customer focus, top management commitment, process management (F4) are among the top prominent TQM CSF's (Al-Khalifa and Aspinwall 2008; Antony et al. 2004; Jamali et al. 2011; Khanna et al. 2011; Rezazadeh et al. 2012; Gupta et al. 2017).

Conferring to the obtained (D-R) values, eight CSFs for successful adoption of TQM implementation initiatives in

organization are namely: Process management (CSF4) > Customer focus (CSF5) > Supplier development (CSF7) > Product and service design (CSF13) > Continuous improvement (CSF12) > Existing organizational culture (CSF10) > Supplier quality management (CSF2) > Employee involvement (CSF8) have been categorized into the effect group CSFs.

Additionally, seven CSF's namely Top management commitment (CSF1) > Training and education (CSF6) > Human resource management (CSF3) > Information system (CSF11) > Employee empowerment (CSF15) > Inter organizational linkage (CSF14) > Team work (CSF9) have been categorized into the cause group.

In the Fig. 5, the correlation among the fifteen different CSFs is presented, which demonstrates that CSF1 exhibits the highest correlation with respect to other presented CSFs; because top management commitment towards TQM

Table 7 The total relation matrix

	CSF1	CSF2	CSF3	CSF4	CSF5	CSF6	CSF7	CSF8
CSF1	[0.346,0.499]	[0.435,0.591]	[0.375,0.53]	[0.508,0.661]	[0.534,0.681]	[0.388,0.548]	[0.494,0.648]	[0.456,0.615]
CSF2	[0.327,0.464]	[0.292,0.431]	[0.269,0.415]	[0.391,0.534]	[0.423,0.562]	[0.295,0.44]	[0.399,0.540]	[0.362,0.508]
CSF3	[0.332,0.484]	[0.341,0.503]	[0.258,0.406]	[0.425,0.58]	[0.421,0.580]	[0.349,0.498]	[0.397,0.558]	[0.378,0.537]
CSF4	[0.348,0.503]	[0.368,0.527]	[0.305,0.463]	[0.381,0.534]	[0.455,0.608]	[0.338,0.495]	[0.430,0.585]	[0.386,0.547]
CSF5	[0.378,0.522]	[0.394,0.541]	[0.311,0.466]	[0.469,0.611]	[0.401,0.547]	[0.345,0.498]	[0.446,0.593]	[0.395,0.551]
CSF6	[0.342,0.505]	[0.374,0.537]	[0.333,0.489]	[0.474,0.625]	[0.464,0.621]	[0.29,0.447]	[0.428,0.590]	[0.405,0.567]
CSF7	[0.356,0.507]	[0.382,0.535]	[0.296,0.452]	[0.46,0.607]	[0.458,0.609]	[0.317,0.479]	[0.367,0.520]	[0.389,0.548]
CSF8	[0.331,0.493]	[0.351,0.516]	[0.313,0.47]	[0.441,0.597]	[0.437,0.598]	[0.311,0.478]	[0.424,0.583]	[0.327,0.487]
CSF9	[0.344,0.505]	[0.382,0.541]	[0.324,0.481]	[0.457,0.612]	[0.470,0.624]	[0.323,0.489]	[0.446,0.601]	[0.418,0.574]
CSF10	[0.332,0.497]	[0.375,0.536]	[0.322,0.480]	[0.461,0.615]	[0.457,0.615]	[0.334,0.496]	[0.449,0.604]	[0.399,0.562]
CSF11	[0.375,0.53]	[0.401,0.558]	[0.341,0.496]	[0.484,0.634]	[0.481,0.636]	[0.342,0.506]	[0.455,0.612]	[0.415,0.576]
CSF12	[0.332,0.483]	[0.357,0.509]	[0.300,0.447]	[0.441,0.584]	[0.449,0.593]	[0.323,0.471]	[0.430,0.574]	[0.386,0.537]
CSF13	[0.346,0.497]	[0.371,0.524]	[0.280,0.438]	[0.447,0.595]	[0.450,0.600]	[0.325,0.481]	[0.420,0.573]	[0.371,0.532]
CSF14	[0.353,0.505]	[0.355,0.516]	[0.295,0.455]	[0.43,0.588]	[0.428,0.589]	[0.31,0.474]	[0.409,0.570]	[0.373,0.537]
CSF15	[0.333,0.484]	[0.347,0.503]	[0.320,0.470]	[0.449,0.596]	[0.434,0.588]	[0.331,0.485]	[0.426,0.577]	[0.4,0.552]
	CSF9	CSF10	CSF11	CSF12	CSF13	CSF14	CSF15	
CSF1	[0.45,0.604]	[0.483,0.634]	[0.432,0.584]	[0.458,0.612]	[0.475,0.624]	[0.417,0.572]	[0.398,0.553]	
CSF2	[0.369,0.503]	[0.389,0.524]	[0.339,0.479]	[0.379,0.513]	[0.385,0.521]	[0.322,0.467]	[0.302,0.444]	
CSF3	[0.384,0.535]	[0.422,0.57]	[0.353,0.506]	[0.39,0.542]	[0.359,0.518]	[0.328,0.488]	[0.356,0.502]	
CSF4	[0.387,0.541]	[0.403,0.561]	[0.355,0.512]	[0.41,0.56]	[0.415,0.565]	[0.336,0.498]	[0.328,0.483]	
CSF5	[0.401,0.548]	[0.424,0.568]	[0.374,0.519]	[0.413,0.556]	[0.431,0.572]	[0.361,0.513]	[0.352,0.498]	
CSF6	[0.394,0.553]	[0.434,0.589]	[0.39,0.542]	[0.424,0.576]	[0.428,0.58]	[0.336,0.504]	[0.358,0.514]	
CSF7	[0.395,0.546]	[0.428,0.576]	[0.357,0.512]	[0.401,0.553]	[0.395,0.546]	[0.345,0.502]	[0.342,0.495]	
CSF8	[0.387,0.543]	[0.415,0.57]	[0.361,0.518]	[0.393,0.55]	[0.391,0.55]	[0.332,0.496]	[0.342,0.497]	
CSF9	[0.336,0.491]	[0.447,0.596]	[0.375,0.531]	[0.391,0.552]	[0.400,0.56]	[0.333,0.501]	[0.354,0.510]	
CSF10	[0.389,0.548]	[0.362,0.517]	[0.367,0.525]	[0.412,0.567]	[0.410,0.567]	[0.348,0.511]	[0.358,0.512]	
CSF11	[0.399,0.558]	[0.439,0.594]	[0.327,0.481]	[0.422,0.578]	[0.433,0.586]	[0.363,0.525]	[0.366,0.522]	
CSF12	[0.376,0.524]	[0.403,0.55]	[0.351,0.495]	[0.333,0.476]	[0.409,0.55]	[0.326,0.477]	[0.341,0.482]	
CSF13	[0.361,0.515]	[0.404,0.556]	[0.341,0.498]	[0.378,0.534]	[0.335,0.484]	[0.329,0.488]	[0.331,0.484]	
CSF14	[0.357,0.519]	[0.394,0.553]	[0.343,0.502]	[0.373,0.534]	[0.384,0.542]	[0.277,0.434]	[0.334,0.489]	
CSF15	[0.384,0.535]	[0.411,0.561]	[0.346,0.497]	[0.395,0.545]	[0.399,0.549]	[0.327,0.487]	[0.289,0.435]	

implementation is necessary quality management practice and their respective concern however also influence other quality management factors and organization process.

The similar work from the literature revealed that findings of Kutlu and Kadaifci (2014) exhibited that by using fuzzy cognitive mapping to analyse critical successful factors for TQM, “Top Management Commitment and Leadership” and “Strategic Planning” were found to be the most influencing drivers of a successful TQM implementation among 9 factors. Georgiev and Ohtaki (2019) showed that among the 12 CSFs that they identified, evidence for the existence of soft constructs have not been considered previously (i.e. middle management involvement and support, reward and recognition, CSR focus). Salaheldin (2009) integrated the CSFs of TQM practices, i.e. strategic, tactical

and operational factors, with operational and organizational performances as related drivers of the effectiveness and success of TQM practices in the SMEs. Antony et al. (2002) revealed that training and education are the most critical factor for the successful implementation of TQM in HK organisations followed by quality data reporting, management commitment.

Furthermore, in the present work the identified fifteen different TQM critical success factors for successful implementation in automobile industry have been mapped and this represents a visual structure to decision-makers. In the applied analysis the ‘prominence’ comprises of factors integration from both an effect (resulting) and cause (influencing) perspective. However, the applied analysis will result, with an ordinal (temporal) view on initial place, what

Table 8 Cause/effect parameters for CSFs for successful adoption of TQM initiatives in the SMES

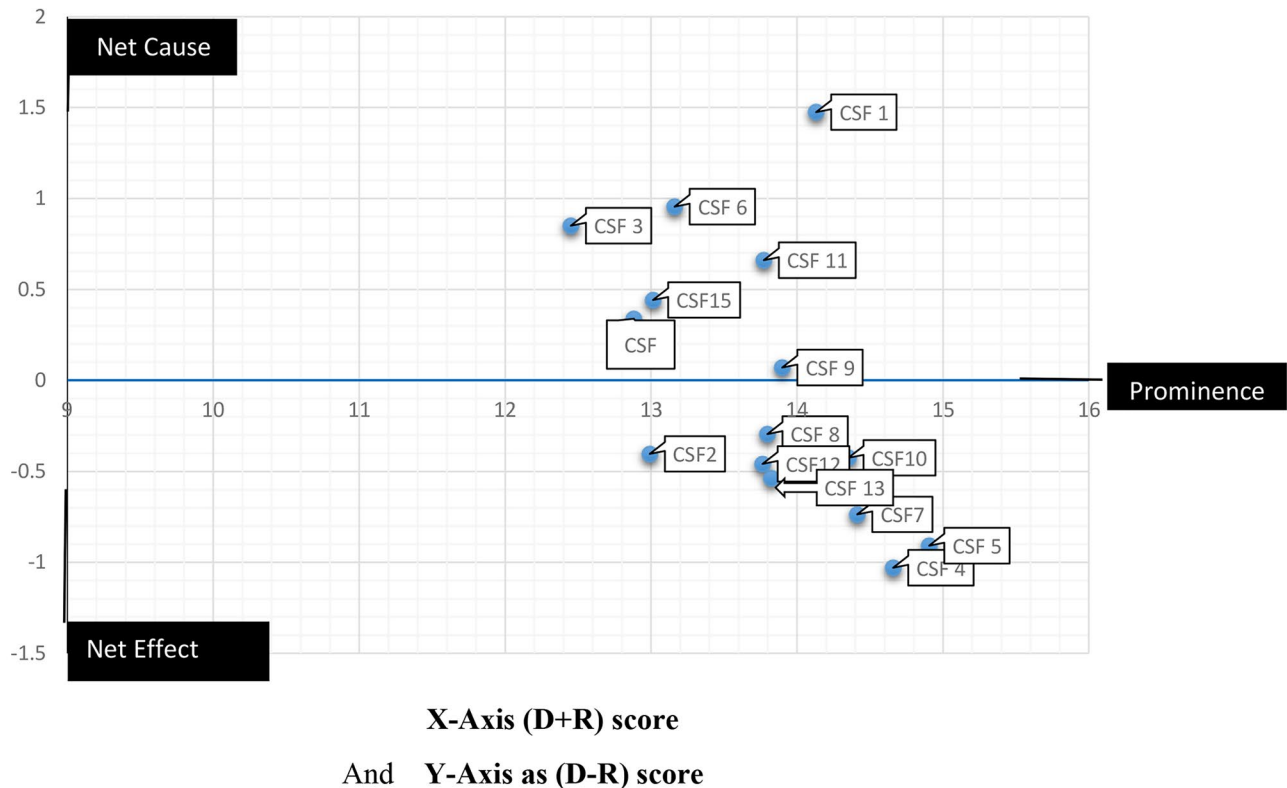
Critical success factor		D	R	Prominence D + R	Net Effect D-R	Cause/ Effect
CSF 1	(TMC)	[6.650,8.956]	[5.175,7.481]	[11.825,16.437]	[-0.831,3.781]	Cause
CSF 2	(SQM)	[5.242,7.346]	[5.526,7.868]	[10.768,15.213]	[-2.626,1.819]	Effect
CSF 3	(HRM)	[5.494,7.808]	[4.642,6.959]	[10.135,14.767]	[-1.465,3.166]	Cause
CSF 4	(PM)	[5.644,7.984]	[6.716,8.972]	[12.362,16.956]	[-3.328,1.268]	Effect
CSF 5	(CF)	[5.895,8.102]	[6.762,9.051]	[12.658,17.153]	[-3.155,1.339]	Effect
CSF 6	(T&E)	[5.876,8.240]	[4.921,7.286]	[10.796,15.526]	[-1.410,3.319]	Cause
CSF 7	(SD)	[5.688,7.988]	[6.421,8.728]	[12.108,16.716]	[-3.043,1.568]	Effect
CSF 8	(EI)	[5.557,7.946]	[5.861,8.230]	[11.418,16.175]	[-2.673,2.085]	Effect
CSF 9	(TW)	[5.810,8.169]	[5.767,8.062]	[11.567,16.232]	[-2.262,2.401]	Cause
CSF 10	(OC)	[5.777,8.152]	[6.257,8.521]	[12.034,16.672]	[-2.744,1.895]	Effect
CSF 11	(IS)	[6.042,8.391]	[5.410,7.701]	[11.453,16.091]	[-1.659,2.986]	Cause
CSF 12	(CI)	[5.553,7.752]	[5.974,8.248]	[11.526,16.001]	[-2.695,1.779]	Effect
CSF 13	(P-Design)	[5.488,7.799]	[6.048,8.313]	[11.536,16.113]	[-2.825,1.751]	Effect
CSF 14	(I-O)	[5.414,7.807]	[5.081,7.463]	[10.494,15.275]	[-2.049,2.726]	Cause
CSF 15	(EE)	[5.591,7.865]	[5.152,7.421]	[10.743,15.286]	[-1.83,2.7132]	Cause

(CSF1, CSF6, CSF3, CSF11, CSF15, CSF14, CSF9)—*Cause group*, and(CSF4, CSF5, CSF7, CSF13, CSF12, CSF10, CSF2, CSF8)—*Effect group*

factors are required (immediate needed customer focus policies), and which factors will need attention at an upcoming time. In the following discussion based on analysis we will categories and evaluate relationship among factors.

6.1 Influential factors

CSFs having maximum interaction influence on other CSFs, refer to have the highest relation and prominence.

**Fig. 4** Overall Prominence of CSFs for successful TQM implementation

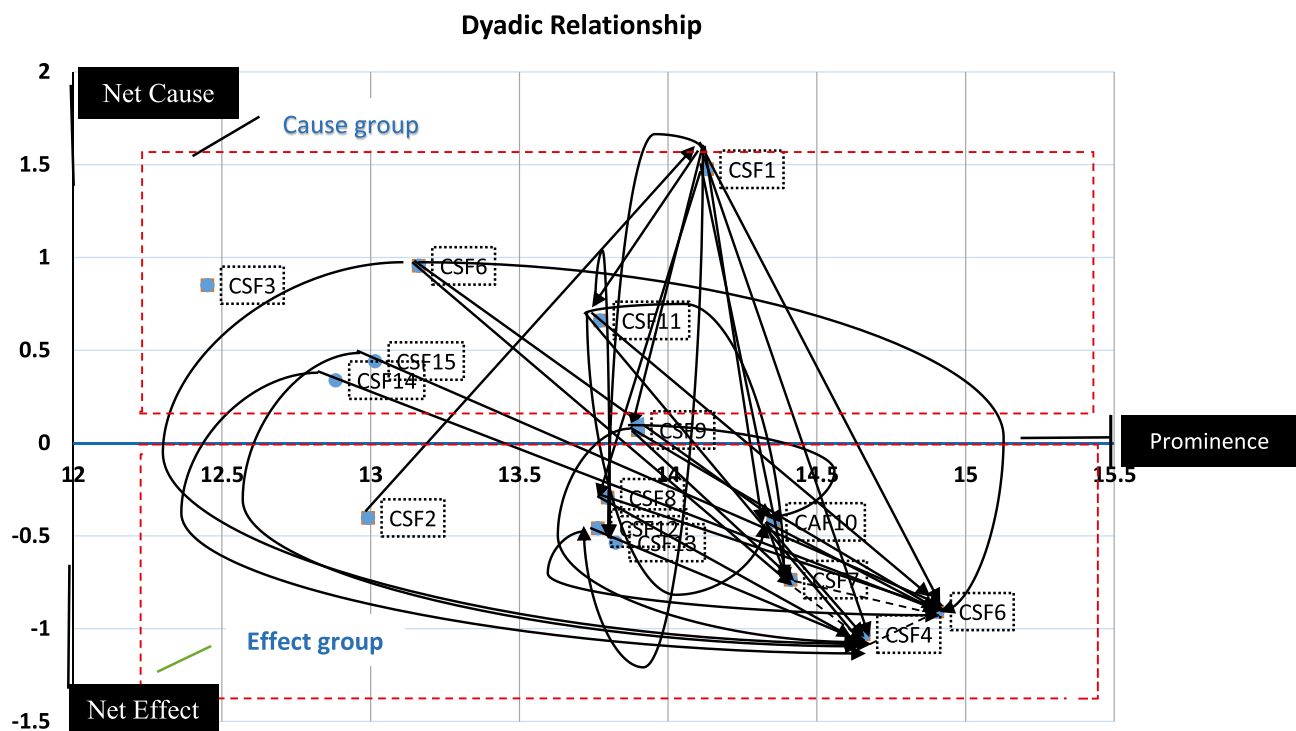


Fig. 5 Overall DEMATEL prominence-causal relationship diagram. (All the relationship that meets or exceeds the threshold value $(\phi)=[0.05014,0.4584]$)

With respect to present research work from Fig. 5, out of 15 identified CSF's seven key net cause CSF namely (CSF1, CSF3, CSF6, CSF9, CSF11, CSF14, CSF15) comes under the influential region with net effect score over '0'. It describes that all these presented seven influential factors not only plays a significant role while incorporating the TQM implementation initiatives in the industry but they can also affect other CSF. These seven influential CSF's also belong to the factors of cause group and have an effect on implementation of TQM in industries. Among them Top management commitment (CSF 1) and Training and education (CSF 6) are mainly two most vital influential CSF, which are primarily related to internal capability having highest (R-D) score.

'**Top management commitment (CSF1)**' has the maximum (D-R) score among the entire cause group CSFs, which mainly indicates that for TQM implementation, CSF 1 has the highest influence on the entire system.

On the other hand, its obtained (D + R) score, which is somewhat low, could be justifiable from the fact that other factors can be easily affected by top management commitment but in return it receive moderately small influence. At the same time having the highest influential impact driver (R) on the overall system plays a very influential role in business organization's processes and quality management initiatives. It has also been seen by numerous researcher's study on TQM implementation that top management commitment

has been a key factor (Abdullah Hokoma et al. (2010); Gupta et al. (2017)); this supports the facts presented in the research work. According to Taylor and Wright (2003), involvement and proper understanding of process in organizations is focused on customer requirements and expectations by the organizations senior managers are quite crucial antecedents of TQM success. Similarly, Ahmed et al. (2005), concluded that strong top management commitment is the determining factor for quality management success. Finding by (Gupta et al. 2017; Suwandej 2015) also support that for effective implementation of TQM, TMC (CSF1) is most instrumental, further the effective leaders possess the strong desire to grow and characteristic of people and management of resources aids in constructing a robust quality management program. In addition, it has been observed that the TQM understanding, its implementations, and its importance are commonly neglected. Consequently, there is a necessity for the top management to be sensitized for the introduction of a robust culture that empowers each employee to devise an efficient method and systems to inculcate continuous quality improvement culture without any interference.

The second top CSF in the (D-R) column is the '**Training and education (CSF6)**', which also has reasonable power to affect other factors as given by its influential impact index (R) value. The obtained results from the study conducted by Cetindere et al. (2015), is complementing present study. According to which for enhancing organization business

performance and also for providing higher quality products and services it's quite important to emphasize on the employees training.

In Indian SMES sector, the adoption of quality management practices is still considered as a stigma.

'**Human resource management (CSF3)**', attains the third rank among TQM CSFs based on its obtained (D-R) score in Table 8. Clearly indicating its prominence, but at the same time having the lowest influential impact driver (R) on the overall system. It signifies that human resource management doesn't have much effect on TQM implementation for the given case study. From Fig. 5, it can be seen that HRM (CSF3) is neither a significant net cause factor nor a net effect factor. In this study managers observe human resource management as a less integrated critical success factor for implementation of TQM. In Indian organizations, this factor lacks its importance, the possible reason behind this may be the misconception that TQM is only related to top management decision directions for quality improvement, and the employees may not concern much about it. In contrast to the finding in Indian organization, HRM has been considered as a less effective CSF. According to Blackburn and Rosen (1995), for the success of TQM, the HR function should construct a blueprint of TQM through which its activities can be guided. Undoubtedly, in the knowledge economy, human capital is the most important capital of organizations (Mehralian et al. 2012). In addition, continuous improvement can be realized if the employee continuously attains new knowledge and actively involve themselves consequently errors can be easily recognized and employee can solve glitches much more efficiently and effectively (Yan et al. 2019).

Further, for product quality-related business gains, practitioners and managers continuously try to achieve total quality management goals. TQM from system perspective requires improving the quality of an organization's outputs, including goods and services, through continual improvement of internal practices. Similarly, succeeding CSF '**Information system (CSF11)**', attains fourth rank among other TQM CSF in Table 8. Similar findings obtained by Khanna et al. 2011 also reveals that information system is one of the top TQM CSF. They emphasized that it's essential to have an effective and efficient system for proper collection of quality-related information such as cost of rework, scrap and quality data. Its use also provides a foundation for a lot of quality-related activities. This allows organizations to collect data in timely manner and share it transparently so that it can be utilized to give employees a proper feedback. In organizations, TQM basically emphasize on ensuring that the outputs of the information system to be used for performance measurement (Khanna et al. 2011). For SMES's performance measurement can aid managers to maintain performance and viability of the firm. This can be done by scanning and

interpretation of environmental changes, as huge uncertainty is associated with the external environment for the new ventures. (Elenkov 1997; Sharma and Bhagwat 2007).

Further, '**Employee empowerment (CSF15)**' in TQM implementation with (D-R) score holds fifth rank. According to Kairiša and Lapiņa (2019), employee empowerment promotes the organization's unity and development, which is the basic characteristic of an "adhocratic culture". According to (Kyoon Yoo et al. 2006), employee empowerment has influence on both internal quality as well as on external quality outcomes.

Lastly, in the cause group, the factors '**Inter-organizational linkages (CSF14)**' and '**Team Work (CSF9)**' attains lowest (D-R) scores as can be visually observed from Figs. 4 and 5 and based on its obtained values in Table 8. It was observed that there are very wide differences of opinion between the quality and production departments on many organization-related matters (Amar and Mohd Zain 2002). Employee relations and coordination between departments influence the performance of the organizational system and consequently determine the nature and extent of TQM implementation. However, teamwork in TQM encompasses collaboration, and between various functions and between non-managers and managers (Dean and Bowen 1994). The working on these factors will assist managers to formulate policies for successful TQM implementation in SMES context.

6.2 Resulting factors

According to Fontela and Gabus 1976, the net effect CSF's represents the most influenced and resulting factors. The factors in these groups are the core problems, having high prominence and low relation. These factors get influenced by other factors effortlessly. From Fig. 5 having a net effect score below '0', from Fig. 5, for the presented study of TQM implementation there are total eight key net effect CSF's. These factors come under resulting factors group. It may also be defined as resulting factors are the factors that possibly will be addressed lastly by industries such that TQM implementation turns out to be successful. But these CSF's are not necessarily having low significance since many of the initial foundational critical success factors result in single resulting CSF's. A resulting CSF may serve as the ultimate gatekeeper for the success of TQM implementation. However, in our study no factor is serving as a resulting factor. Figure 5, shows that the direction of significant resulting factor among customer focus and satisfaction (CSF5), process management (CSF4) and supplier development (CSF7) is unidirectional.

In effect group '**Employee Involvement (CSF 8)**' in (D-R) score column holds highest position among other resulting factors. The participation of employee's in the organization quality program results in increased information and

knowledge flow to the organization's end level for resolving problems (Snell and Dean 1992).

The next second highest CSF in effect group is '**Supplier quality management (CSF 2)**' as per (D-R) score column in Table 8. The quality product production involves incoming materials with good quality. Thus, it's essential for each selected supplier for TQM implementation that it should also be trained for quality (Khanna et al. 2011). Subsequently long-term, co-operative relationships need to be maintained with limited number of suppliers instead of several ones so that both parties' relationships are mutually beneficial. To check the incoming material quality and supplier delivery performance system must be required to get ensured about quality-related issues.

In (D-R) column '**Existing organizational culture (CSF 10)**' obtains third position as resulting factor. In organization the spread of quality philosophy is accelerated by means of orientation of strong quality (Maletič et al. 2014). In contrast, Welikala And and Sohal 2008 commented that a lack of quality culture possibly will obstruct the problem solving through the organization and as a result it obstructs continuous improvement.

In our finding for TQM implementation '**Process management (CSF4)**' comes under the effect group. It acts as a strong tool through which quality assurance can be enhanced via proper periodic review at each manufacturing operation and consistent corrective actions aspect (Khanna et al. 2011). **Process management (CSF 4) and Customer focus & Satisfaction (CSF5)**, as per the obtained (D-R) score of respectively positioned at last in effect group. Figure 5, shows that **Process management (CSF 4) and Customer focus & satisfaction (CSF 5)** get highly influenced by the cause group factors and also direction in between them is bi-directional. This means that change in process management is highly based on satisfaction level of customer. To achieve this, the processes should be made flexible. In addition, Raj and Attri (2011) stated that in organization, TQM provides an umbrella under that every person try and develop customer satisfaction at persistently lower real costs. Quality improvement is aided by process management by use of automated testing, employee's self-inspection, preventive maintenance, foolproof process design, selective automation and statistical process control (Kr Singh 2011). As earlier stated, some research works previously, reported that implementation of the TQM in organization has a positive impact on customer satisfaction (Kumar et al. 2009; Wang et al. 2012).

Furthermore, the other TQM implementation factors that follow the sequence of increasing order of priority list in the effect group, include Continuous improvement (CSF 12), Product and service design (CSF13), and Supplier development (CSF 7) respectively. '**Employee**

Involvement (CSF8)' obtained the highest (R-D) score in effect group, which suggests that this factor receives the least impact. The factor 'CSF5' and 'CSF 4' according to its attained (D + R) score of are the most significant factor among other CSF's. Organizational output can be continually improved by performance improvement (Corbett and Rastrick 2000; Yan et al. 2019).

Notably, all experts involved in this study showed an agreement with the results. Since the user data and analysis are predominantly based on industry operating in India. From findings we can't generalize its applicability for other industries or similar sectors engaged in manufacturing and production process. It is expected to provide the managers an insight into the quality related aspects from the presented framework and its findings. Depending upon the specific industry the applied framework can be utilized with minimal modifications to establish the causal relationship among the identified CSFs for TQM implementation in their respective industries.

7 Implication to theory and practices

7.1 Implication to theory

This research work and its findings will contribute theoretically to the existing literature as:

- Provides a detailed Grey based DEMATEL procedure to categorize and reveal a relationship among the identified CSF's.
- Categorizing the TQM CSF's into cause-and-effect group so that the practicing managers and industry experts can focus their effort on implementing TQM successfully.
- Providing guidelines to industry managers in developing long term strategies in achieving the sustainable supply chain.
- Pointing towards a lack of research for successful TQM adopting in a competitive environment.
- Identifying the most influential CSF (Top management commitment under this study) among recognized CSFs for the industry. Identifying the most resulting and influential factors through an exhaustive literature review.
- Identification of prominence among i.e. the ordinal (temporal) view on in initial place what factors are required (immediate needed customer focus policies), and which factors will need attention at an upcoming time.
- Article presented the correlation among the fifteen different CSFs.
- This article makes an important contribution by structuring the complex research field and by providing advice for TQM implementation as well as for further research.

7.2 Implication to practices

This research work will give insight and assist planners and practice managers in the SMES, to identify the most influential (cause) and resulting (effect) group CSFs for successful TQM implementation. Global competitive environment, pressurizing each industry to form activities and policies which combines people, process & technology in an institutional infrastructure that provides vision incentive, support to organization for achieving the goal of total quality.

Identifying, analyzing, prioritizing and finding the causal interactions among several critical success factors are essential for the goal to successfully initiative to implementation of TQM for the particular industry. The categorization of identified CSF's may assist practice managers decision making during implementing TQM initiatives in the respective industry.

The findings show that managers must first emphasize on 'customer focus, and process management' on developing guidelines because these factors attain high prominence w.r.t other CSF's. Secondly, they must focus on ensuring top management commitment w.r.t process and teams since it's the highest influential factor in cause group. The findings in this study also identified different TQM critical success factors for successful implementation in automobile industry that have been mapped and this represents a visual structure to decision-makers.

Supplier development and existing organizational culture are also other important CSFs in the list, that experts and managers need to focus on the subject industry. Managers employed in the SMES can make use of the results obtained in study to adopt TQM implementation initiatives, which will not only assist industry in achieving competitive goal but it also lessens the total costs associated to poor quality, that overall results in improved market value and industry image globally.

8 Conclusion and directions for further research

In the present study out of fifteen identified CSF's seven influential factors plays a significant role in incorporating the TQM implementation initiatives in the industry. These seven influential CSF's also belong to the factors of cause group and have an effect on implementation of TQM in industries. Among them Top management commitment, Training and education, and Human resource management are the three most vital influential CSF's. The realization of TQM implementation in an organization can be accomplished by these two aspects: First, by managing the human resources by providing unvarying training and education necessary at all the levels in the organization; secondly, the adherence of the top management towards its commitment for achieving the total quality goal in all the internal as well as external activities.

In this study a Grey based DEMATEL framework has been successfully established in order to investigate the interrelation among identified CSFs and to categorize them for successful TQM implementation in different organizations. Depending on industry sector, their priorities, size, region in which they are located and variation of expert's opinion regarding factor evaluation, the accomplished findings may differ with the obtained results. Future research could scrutinize the findings selecting a large and different sample set. The applied methodology may be extended for the purpose to improve industry performance in different sectors. This study can be extended by including different evolving factors that may influence TQM implementation and other decision-making models can be applied like ISM-MICMAC, Fuzzy-DEMATEL to find the interaction and prioritization among the different TQM CSFs. Future research may be conducted in the context of other industries situated and operating in developed/developing countries to compare the findings with this study. Lastly, TQM implementation is evolving into a complex practice, the number factors affecting it and the key elements are found to increase.

Data availability Data (expert opinion) obtained from four Industry and Academia experts are available and can be provided.

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